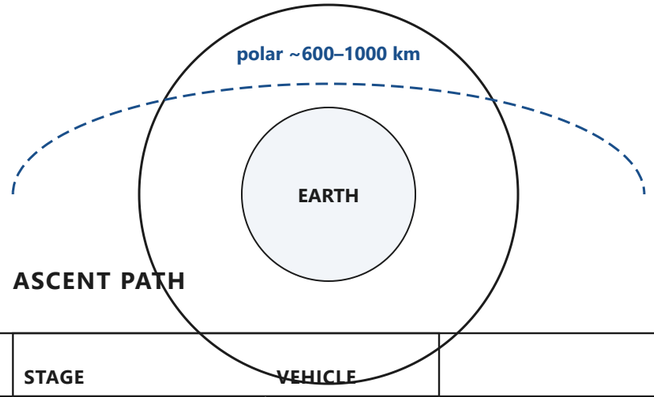
THE INSIGHT

— When the mesh needs backhaul beyond terrestrial reach, our own store-and-forward payload flies.
— Nobody else launching smallsats has a deployed rural ground network waiting — the mesh IS the user-terminal layer.
— A text message or sensor packet needs COVERAGE, not Starlink bandwidth — a handful of CubeSats can blanket Canada.
— Higher orbit is correct for THIS layer: bigger footprint per satellite = fewer needed.

ORBIT CONCEPT



PAYLOAD — DERIVED FROM THE MESH

ELEMENT	IMPLEMENTATION
Radio	UHF/S-band store-and-forward transceiver — LoRa-derived or CC1101-class
Envelope	same signed mesh-net format — satellite is just another hop
Store	onboard flash — holds envelopes until ground station passes
Bus	1U/3U CubeSat (GOMspace/EnduroSat/CubeSat Shop) or smallsat
Ground segment	the deployed mesh + community gateways = user terminals

ASCENT PATH

STAGE	VEHICLE	COST CLASS
Hosted payload	existing Canadian sat	$50–500k
University CubeSat	CSA CubeSat Initiative	$50–500k
Dedicated smallsat	rideshare (non-US ok)	$1–5M
Pathfinder	6–12 sats, own ops	$10–30M
Constellation	regional Canada	earned by demand

inclination ~89° — Arctic coverage

MISSION PROFILE

— Polar LEO constellation blankets Canada incl. North — messaging/sensor tier.
— Latency: store-and-forward hours — right-sized for alerts, notices, telemetry.
— Each sat serves hundreds of nodes per pass — plenty for text/sensor traffic.
— Pathfinder: 2–6 sats prove the constellation, then scale regional.

FUNDING PATH

— CSA CubeSat Initiative — funds Canadian university smallsats.
— Provincial/territorial + Indigenous services — northern mandate.
— IDEaS/defence dual-use — CAF buys commercial satcom.
— Strategic: demand-side anchor for domestic launch — the summit.

HONEST STATUS

— Vision stage — no satellite exists. The mesh must prove itself first (L1→swarm→revenue).
— This sheet is mission concept, not spacecraft design — payload inherits mesh crypto/routing.
— Every earlier rung funds and justifies this one — the trailhead is the ground network.

ARCTICCHAIN — RURAL NODE FLEET

DRAWN BY
Surafel Negussie
CHECKED BY
DATE
PROJECT

SCALE
NTS **1/1**
SHEET
DWG NO **AF-H-101**
UNITS mm

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